Chapter 8. Closed Forms and Exact Forms

The Poincaré lemma and the de Rham groups of punctured euclidean space

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Overview

In physics one often asks whether a vector field F in $\mathbb{R}^3$ is the gradient of a scalar field f (F is **conservative**, and f, or its negative, is a **potential function** for F): in the language of forms, whether a 1-form is the differential of a 0-form, i.e. exact. Similarly one asks whether G is the curl of a field F: whether a 2-form is exact. In $\mathbb{R}^n$: if ω is a k-form on an open set A, a necessary condition for exactness is $d\omega = 0$. It is not in general sufficient; this chapter explores what additional conditions on A, or on A and ω, ensure that ω is exact.

§39. The Poincaré Lemma

Motivation. Every closed form is locally exact, and on star-shaped domains it is globally exact. This settles when a vector field has a potential in a convex region, and it is the local fact whose global failure is measured by de Rham cohomology.

If A is *star-convex*, every closed form on A is exact (the Poincaré lemma). We begin with a preliminary result.

Theorem 39.1 (Leibniz's rule)

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \times [a,b] \to \mathbb{R}$ be a continuous function. Denote f by $f(\mathbf{x}, t)$ for $\mathbf{x} \in Q$ and $t \in [a,b]$. Then the function

$$F(\mathbf{x}) = \int_{t=a}^{t=b} f(\mathbf{x}, t)$$

is continuous on Q. Furthermore, if $\partial f/\partial x_j$ is continuous on $Q \times [a,b]$, then

$$\frac{\partial F}{\partial x_j}(\mathbf{x}) = \int_{t=a}^{t=b} \frac{\partial f}{\partial x_j}(\mathbf{x}, t).$$

This is *Leibniz's rule for differentiating under the integral sign.*

Proof.

Step 1. $Q \times [a,b]$ is compact, so f is uniformly continuous: given $\epsilon > 0$ there is $\delta > 0$ with $|f(\mathbf{x}_1, t_1) - f(\mathbf{x}_0, t_0)| < \epsilon$ whenever $|(\mathbf{x}_1, t_1) - (\mathbf{x}_0, t_0)| < \delta$. So for $|\mathbf{x}_1 - \mathbf{x}_0| < \delta$,

$$|F(\mathbf{x}_1) - F(\mathbf{x}_0)| \le \int_{t=a}^{t=b} |f(\mathbf{x}_1, t) - f(\mathbf{x}_0, t)| \le \epsilon(b-a).$$

Step 2. Only x_j and t are involved, so we may assume $n = 1$ and $Q = [c,d]$. Let $G(x) = \int_{t=a}^{t=b} D_1 f(x,t)$; we show $F'(x) = G(x)$. Since $D_1 f$ is continuous on $[c,d] \times [a,b]$, Fubini's theorem and the fundamental theorem of calculus give, for $x_0 \in [c,d]$,

$$\int_{x=c}^{x=x_0} G(x) = \int_{x=c}^{x=x_0} \int_{t=a}^{t=b} D_1 f(x,t) = \int_{t=a}^{t=b} \int_{x=c}^{x=x_0} D_1 f(x,t) = \int_{t=a}^{t=b} [f(x_0,t) - f(c,t)] = F(x_0) - F(c).$$

G is continuous by Step 1, so the fundamental theorem of calculus gives $G(x) = F'(x)$. ∎

Definition 39.1 (differentiable homotopy)

Let A, B be open in $\mathbb{R}^n$, $\mathbb{R}^m$, and $g, h : A \to B$ of class C^∞. g and h are **differentiably**

homotopic if there is a C^∞ map $H : A \times I \to B$ ($I = [0,1]$) with

$$H(\mathbf{x}, 0) = g(\mathbf{x}) \qquad \text{and} \qquad H(\mathbf{x}, 1) = h(\mathbf{x}) \qquad (\mathbf{x} \in A).$$

H is a **differentiable homotopy** between g and h.

For each t, $\mathbf{x} \mapsto H(\mathbf{x}, t)$ is a C^∞ map $A \to B$; thinking of t as time, H "deforms" g into h as t goes from 0 to 1.

Theorem 39.2

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g, h : A \to B$ be C^∞ maps that are differentiably homotopic. Then there is a linear transformation

$$\mathcal{P} : \Omega^{k+1}(B) \to \Omega^k(A),$$

defined for $k \geq 0$, such that for any form η of order $k > 0$,

$$d\mathcal{P}\eta + \mathcal{P}d\eta = h^*\eta - g^*\eta,$$

while for a form f of order 0,

$$\mathcal{P}df = h^*f - g^*f.$$

Hence if η is a closed form of positive order, $h^*\eta$ and $g^*\eta$ differ by an exact form ($= d\mathcal{P}\eta$); if f is a closed 0-form, $h^*f = g^*f$. (d raises order by 1 and $\mathcal{P}$ lowers it by 1; $\mathcal{P}f$ is undefined for a 0-form f.)

Proof.

Step 1. A special case. Given A open in $\mathbb{R}^n$, let U be a neighborhood of $A \times I$ in $\mathbb{R}^{n+1}$ and $\alpha, \beta : A \to U$, $\alpha(\mathbf{x}) = (\mathbf{x}, 0)$, $\beta(\mathbf{x}) = (\mathbf{x}, 1)$ (differentiably homotopic). We define for each $(k+1)$-form η on U a k-form $P\eta$ on A with

$$dP\eta + Pd\eta = \beta^*\eta - \alpha^*\eta \ \ (\text{order } \eta > 0), \qquad Pdf = \beta^*f - \alpha^*f \ \ (\text{order } f = 0). \qquad (*)$$

Let $\mathbf{x}$ be the general point of $\mathbb{R}^n$ and t that of $\mathbb{R}$; $dx_1, \ldots, dx_n, dt$ are the elementary 1-forms of $\mathbb{R}^{n+1}$. For a continuous scalar function g on $A \times I$ define $(\mathcal{I}g)(\mathbf{x}) = \int_{t=0}^{t=1} g(\mathbf{x}, t)$. For $k \geq 0$ write a $(k+1)$-form uniquely as

$$\eta = \sum_{[I]} f_I \, dx_I + \sum_{[J]} g_J \, dx_J \wedge dt,$$

I ascending $(k+1)$-tuples and J ascending k-tuples from $\{1, \ldots, n\}$, and define

$$P\eta = \sum_{[I]} P(f_I \, dx_I) + \sum_{[J]} P(g_J \, dx_J \wedge dt), \qquad P(f_I \, dx_I) = 0, \quad P(g_J \, dx_J \wedge dt) = (-1)^k (\mathcal{I}g_J) \, dx_J.$$

$P\eta$ is a k-form on A; P is linear (uniqueness of the representation and linearity of $\mathcal{I}$), and $\mathcal{I}g$ is C^∞ by Leibniz's rule, so $P\eta$ is C^∞. For $k = 0$, $\eta = \sum f_i \, dx_i + g \, dt$, J is empty, and $P\eta = \mathcal{I}g$. (P "integrates η in the direction of the last coordinate".)

Step 2. The formulas $P(f \, dx_I) = 0$ and $P(g \, dx_J \wedge dt) = (-1)^k (\mathcal{I}g) \, dx_J$ hold even for arbitrary (non-ascending) I, J: trivially if indices repeat (then $dx_I = 0$, $dx_J = 0$); by definition if ascending; and in general since rearranging indices changes dx_I, dx_J only by a sign.

Step 3. $(*)$ for $k = 0$:

$$P(df) = P\Big(\sum_j \frac{\partial f}{\partial x_j} dx_j\Big) + P\Big(\frac{\partial f}{\partial t} dt\Big) = 0 + (-1)^0 \mathcal{I}\Big(\frac{\partial f}{\partial t}\Big) = f \circ \beta - f \circ \alpha = \beta^* f - \alpha^* f,$$

by the fundamental theorem of calculus.

Step 4. $(*)$ for $k>0$. Since $\alpha(\mathbf{x})=(\mathbf{x},0)$, $\alpha^*(dx_i)=dx_i$ and $\alpha^*(dt)=d(0)=0$; likewise for β. By linearity it suffices to treat $\eta=f\,dx_I$ and $\eta=g\,dx_J\wedge dt$. For $\eta=f\,dx_I$: the left side is

$$dP\eta+Pd\eta=d(0)+P\Big(\sum_j\frac{\partial f}{\partial x_j}dx_j\wedge dx_I\Big)+P\Big(\frac{\partial f}{\partial t}dt\wedge dx_I\Big)=0+(-1)^{k+1}P\Big(\frac{\partial f}{\partial t}dx_I\wedge dt\Big)=\mathcal{I}\Big(\frac{\partial f}{\partial t}\Big)dx_I=[f\circ\beta-f\circ\alpha$$

using Step 2 ($dt\wedge dx_I=(-1)^{k+1}dx_I\wedge dt$ as dx_I has order $k+1$), and the right side is $\beta^*\eta-\alpha^*\eta=(f\circ\beta)\beta^*(dx_I)-(f\circ\alpha)\alpha^*(dx_I)=[f\circ\beta-f\circ\alpha]\,dx_I$. For $\eta=g\,dx_J\wedge dt$:

$$d(P\eta)=d[(-1)^k(\mathcal{I}g)\,dx_J]=(-1)^k\sum_{j=1}^n D_j(\mathcal{I}g)\,dx_j\wedge dx_J. \qquad (**)$$

On the other hand $d\eta=\sum_j(D_jg)\,dx_j\wedge dx_J\wedge dt+(D_{n+1}g)\,dt\wedge dx_J\wedge dt$, and the last term vanishes, so by Step 2

$$P(d\eta)=(-1)^{k+1}\sum_{j=1}^n\mathcal{I}(D_jg)\,dx_j\wedge dx_J. \qquad (***)$$

By Leibniz's rule $D_j(\mathcal{I}g)=\mathcal{I}(D_jg)$, so $(**)+(***)=0$. The right side is $\beta^*(g\,dx_J\wedge dt)-\alpha^*(g\,dx_J\wedge dt)=0$, since $\beta^*(dt)=\alpha^*(dt)=0$.

Step 5. General case. Let $H:A\times I\to B$ be a differentiable homotopy between g and h, $\alpha,\beta:A\to A\times I$ the maps of Step 1, and P the corresponding transformation. Define $\mathcal{P}\eta=P(H^*\eta)$; since $H^*\eta$ is a $(k+1)$-form on a neighborhood of $A\times I$, $\mathcal{P}\eta$ is a k-form on A. As $H\circ\alpha=g$ and $H\circ\beta=h$, for $k>0$

$$d\mathcal{P}\eta+\mathcal{P}d\eta=dP(H^*\eta)+P(H^*d\eta)=dP(H^*\eta)+P(dH^*\eta)=\beta^*(H^*\eta)-\alpha^*(H^*\eta)=h^*\eta-g^*\eta,$$

by Step 1; the case $k=0$ is similar. ■

Definition 39.2 (star-convex)

An open set A in $\mathbb{R}^n$ is **star-convex** with respect to the point $\mathbf{p}$ of A if for each $\mathbf{x}\in A$ the line segment joining $\mathbf{x}$ and $\mathbf{p}$ lies in A.

Example 39.1

A star-shaped region may be star-convex with respect to one of its points $\mathbf{p}$ but not with respect to another point $\mathbf{q}$. A convex set is star-convex with respect to each of its points. A "C"-shaped region is not star-convex with respect to any of its points.

Theorem 39.3 (The Poincaré lemma)

Let A be a star-convex open set in $\mathbb{R}^n$. If ω is a closed k-form on A, then ω is exact on A.

Proof. Let A be star-convex with respect to $\mathbf{p}$. Let $h:A\to A$ be the identity and $g:A\to A$ the constant map to $\mathbf{p}$. They are differentiably homotopic via the **straight-line homotopy**

$$H(\mathbf{x},t)=t\,h(\mathbf{x})+(1-t)\,g(\mathbf{x}),$$

which maps $A\times I$ into A, since $H(\mathbf{x},t)$ lies on the segment from $h(\mathbf{x})=\mathbf{x}$ to $g(\mathbf{x})=\mathbf{p}$. Let $\mathcal{P}$ be as in Theorem 39.2. If f is a 0-form on A, $\mathcal{P}(df)=h^*f-g^*f=f\circ h-f\circ g$; if $df=0$ then $0=f(\mathbf{x})-f(\mathbf{p})$ for all $\mathbf{x}$, so f is constant on A. If ω has order $k>0$, $d\mathcal{P}\omega+\mathcal{P}d\omega=h^*\omega-g^*\omega$, where $h^*\omega=\omega$ (h is the identity) and $g^*\omega=0$ (g is constant). If $d\omega=0$, then $d\mathcal{P}\omega=\omega$, so ω is exact. ■

Theorem 39.4

Let A be a star-convex open set in $\mathbb{R}^n$. Let ω be a closed k-form on A. If $k > 1$, and if η and η_0 are two $k-1$ forms on A with $d\eta = \omega = d\eta_0$, then $\eta = \eta_0 + d\theta$ for some $k-2$ form θ on A. If $k = 1$, and if f and f_0 are two 0-forms on A with $df = \omega = df_0$, then $f = f_0 + c$ for some constant c.

Proof. $d(\eta - \eta_0) = 0$, so $\eta - \eta_0$ is a closed form on A, exact by the Poincaré lemma. Similarly $f - f_0$ is a closed 0-form, hence constant. ■

§40. The deRham Groups of Punctured Euclidean Space

Star-convex open sets are homologically trivial in all dimensions. The simplest set that is not is the punctured euclidean space $\mathbb{R}^n - \mathbf{0}$: earlier we found a closed $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ that is not exact. We now give a definitive criterion for when a closed form on $\mathbb{R}^n - \mathbf{0}$ is exact, using vector spaces $H^k(A)$ called the *deRham groups* of A: A is homologically trivial in dimension k iff $H^k(A)$ is trivial.

Definition 40.1 (quotient space)

Let W be a linear subspace of a vector space V. V/W is the set of subsets of V of the form $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} : \mathbf{w} \in W\}$, called **cosets** of V determined by W. If $\mathbf{v}_1 - \mathbf{v}_2 \in W$ then $\mathbf{v}_1 + W = \mathbf{v}_2 + W$, and if $\mathbf{v}_1 - \mathbf{v}_2 \notin W$ they are disjoint; so V/W is a **partition** of V. With

$$(\mathbf{v}_1 + W) + (\mathbf{v}_2 + W) = (\mathbf{v}_1 + \mathbf{v}_2) + W, \qquad c(\mathbf{v} + W) = (c\mathbf{v}) + W,$$

V/W is a vector space, the **quotient space** of V by W.

The operations are well-defined: if $\mathbf{v}_1 + W = \mathbf{v}_1' + W$ and $\mathbf{v}_2 + W = \mathbf{v}_2' + W$, then $\mathbf{v}_1 - \mathbf{v}_1'$ and $\mathbf{v}_2 - \mathbf{v}_2'$ lie in W, hence so does $(\mathbf{v}_1 + \mathbf{v}_2) - (\mathbf{v}_1' + \mathbf{v}_2')$; similarly for scalar multiplication. If V is finite-dimensional so is V/W; but V/W may be finite-dimensional even when V and W are not.

Definition 40.2 (induced map of quotients)

If $W \subset V$ and $W' \subset V'$ are linear subspaces and $T : V \to V'$ is linear with $T(W) \subset W'$, then $\tilde{T} : V/W \to V'/W'$, $\tilde{T}(\mathbf{v} + W) = T(\mathbf{v}) + W'$, is a well-defined linear transformation, **induced** by T.

Definition 40.3 (deRham group)

Let A be open in $\mathbb{R}^n$. The closed k-forms $\mathcal{C}^k(A)$ and the exact k-forms $\mathcal{E}^k(A)$ are linear subspaces of $\Omega^k(A)$, and $\mathcal{E}^k(A) \subset \mathcal{C}^k(A)$. The **deRham group** of A in dimension k is the quotient vector space

$$H^k(A) = \mathcal{C}^k(A)/\mathcal{E}^k(A).$$

For a closed k-form ω we denote its coset $\omega + \mathcal{E}^k(A)$ by $\{\omega\}$.

$H^k(A)$ is the trivial (zero) vector space iff A is homologically trivial in dimension k. If $g : A \to B$ is a C^∞ map of open sets in $\mathbb{R}^n, \mathbb{R}^m$, then $g^* : \Omega^k(B) \to \Omega^k(A)$ commutes with d, so it carries closed forms to closed forms and exact forms to exact forms, and induces a linear map $g^* : H^k(B) \to H^k(A)$ (we write g^* instead of $\tilde{g}^*$). Two tools from algebraic topology compute these groups: *homotopy equivalence*, and a special case of the *Mayer–Vietoris theorem*.

Theorem 40.1 (Homotopy equivalence theorem)

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g : A \to B$ and $h : B \to A$ be C^∞ maps. If $g \circ h : B \to B$ is differentiably homotopic to the identity map i_B of B, and if $h \circ g : A \to A$ is differentiably homotopic to the identity map i_A of A, then g^* and h^* are linear isomorphisms of the deRham groups.

If $g \circ h = i_B$ and $h \circ g = i_A$, g and h are diffeomorphisms; under the weaker hypotheses of the theorem they are called (differentiable) **homotopy equivalences**.

Proof. If η is a closed k-form on A ($k \geq 0$), Theorem 39.2 implies $(h \circ g)^*\eta - (i_A)^*\eta$ is exact (for $k = 0$ it is zero). So the induced maps satisfy $g^*(h^*(\{\eta\})) = \{\eta\}$: $g^* \circ h^*$ is the identity of $H^k(A)$. Similarly $h^* \circ g^*$ is the identity of $H^k(B)$. The first fact shows g^* maps $H^k(B)$ onto $H^k(A)$ (since $\{\eta\} = g^*(h^*\{\eta\})$); the second shows g^* is one-to-one (if $g^*\{\omega\} = 0$ then $\{\omega\} = h^*(g^*\{\omega\}) = 0$). By symmetry h^* is also a linear isomorphism. ■

Lemma 40.2

Let U and V be open sets in $\mathbb{R}^n$; let $X = U \cup V$; and suppose $A = U \cap V$ is non-empty. Then there exists a C^∞ function $\phi : X \to [0, 1]$ such that ϕ is identically 0 in a neighborhood of $U - A$ and ϕ is identically 1 in a neighborhood of $V - A$.

Proof. Let $\{\phi_i\}$ be a partition of unity on X dominated by $\{U, V\}$, $S_i = \operatorname{Support} \phi_i$. Divide the indices into disjoint sets J and K such that $S_i \subset U$ for $i \in J$ and $S_i \subset V$ for $i \in K$ (e.g. $J = \{i : S_i \subset U\}$, K the rest). Let $\phi(\mathbf{x}) = \sum_{i \in K} \phi_i(\mathbf{x})$; by local finiteness ϕ is C^∞ on X. Let $\mathbf{a} \in U - A$. Choose a neighborhood W of $\mathbf{a}$ meeting only finitely many S_i; let D be the union of those among them with $i \in K$. D is closed and does not contain $\mathbf{a}$ (as $S_i \subset V$ for $i \in K$ and $\mathbf{a} \notin V$), so $W - D$ is a neighborhood of $\mathbf{a}$ on which each ϕ_i, $i \in K$, vanishes; thus $\phi = 0$ on $W - D$. Since $1 - \phi(\mathbf{x}) = \sum_{i \in J} \phi_i(\mathbf{x})$, symmetry shows $1 - \phi$ vanishes in a neighborhood of each point of $V - A$. ■

Theorem 40.3 (Mayer–Vietoris—special case)

Let U and V be open sets in $\mathbb{R}^n$ with U and V homologically trivial in all dimensions. Let $X = U \cup V$; suppose $A = U \cap V$ is non-empty. Then $H^0(X)$ is trivial, and for $k \geq 0$, the space $H^{k+1}(X)$ is linearly isomorphic to the space $H^k(A)$.

Proof. For open $B \subset C$ and a k-form η on C, write $\eta|B = j^*\eta$ ($j : B \to C$ the inclusion), the **restriction** of η to B. Since j^* commutes with d, restrictions of closed (exact) forms are closed (exact); and $(\eta|B)|A = \eta|A$ for $A \subset B \subset C$.

Step 1. $H^0(X)$ is trivial. Let f be a closed 0-form on X. $f|U$ and $f|V$ are closed, and U, V are homologically trivial in dimension 0, so $f|U = c_1$, $f|V = c_2$ are constants; as $U \cap V \neq \emptyset$, $c_1 = c_2$ and f is constant.

Step 2. Let $\phi : X \to [0, 1]$ be C^∞ with $\phi = 0$ in a neighborhood U' of $U - A$ and $1 - \phi = 0$ in a neighborhood V' of $V - A$ (Lemma 40.2). For $k \geq 0$ define $\delta : \Omega^k(A) \to \Omega^{k+1}(X)$ by

$$\delta(\omega) = \begin{cases} d\phi \wedge \omega & \text{on } A, \\ 0 & \text{on } U' \cup V'. \end{cases}$$

Since $d\phi = 0$ on $U' \cup V'$, $\delta(\omega)$ is well-defined, and since A and $U' \cup V'$ are open with union X, it is C^∞ on X. δ is linear and commutes with d up to sign: $d(\delta(\omega)) = -d\phi \wedge d\omega$ on A and 0 on $U' \cup V'$, i.e. $d\delta(\omega) = -\delta(d\omega)$. So δ carries closed forms to closed forms and exact forms to exact forms ($\delta(d\theta) = -d\delta(\theta)$), and induces a linear map $\tilde{\delta} : H^k(A) \to H^{k+1}(X)$. We show $\tilde{\delta}$ is an isomorphism.

Step 3. $\tilde{\delta}$ is one-to-one: if ω is a closed k-form on A with $\delta(\omega) = d\theta$ for a k-form θ on X, then ω is exact. Define k-forms

$$\omega_1 = \begin{cases} \phi\omega & \text{on } A, \\ 0 & \text{on } U', \end{cases} \qquad \omega_2 = \begin{cases} (1-\phi)\omega & \text{on } A, \\ 0 & \text{on } V', \end{cases}$$

well-defined and C^∞ on U and V. Since $d\omega = 0$, $d\omega_1 = d\phi \wedge \omega = \delta(\omega)$ on A and 0 on U', so $d\omega_1 = \delta(\omega)|U = d\theta|U$; thus $\omega_1 - \theta|U$ is closed on U. Similarly $d\omega_2 = -d\theta|V$ and $\omega_2 + \theta|V$ is closed on V. If $k > 0$, homological triviality gives $(k-1)$-forms η_1, η_2 on U, V with $\omega_1 - \theta|U = d\eta_1$, $\omega_2 + \theta|V = d\eta_2$. Restricting to A and adding,

$$\phi\omega + (1-\phi)\omega = \omega_1|A + \omega_2|A = d(\eta_1|A + \eta_2|A),$$

so ω is exact on A. If $k = 0$, there are constants c_1, c_2 with $\omega_1 - \theta|U = c_1$, $\omega_2 + \theta|V = c_2$, and $\phi\omega + (1-\phi)\omega = c_1 + c_2$ on A; so ω is constant, i.e. exact.

Step 4. $\tilde{\delta}$ is onto: given a closed $(k+1)$-form η on X, we find a closed k-form ω on A with $\eta - \delta(\omega)$ exact. $\eta|U$ and $\eta|V$ are closed, so there are k-forms θ_1, θ_2 on U, V with $d\theta_1 = \eta|U$, $d\theta_2 = \eta|V$. Let $\omega = \theta_1|A - \theta_2|A$; it is closed since $d\omega = \eta|A - \eta|A = 0$. Define a k-form θ on X by

$$\theta = \begin{cases} (1-\phi)\theta_1 + \phi\theta_2 & \text{on } A, \\ \theta_1 & \text{on } U', \\ \theta_2 & \text{on } V', \end{cases}$$

well-defined and C^∞. We show $d\theta = \eta - \delta(\omega)$. On A:

$$d\theta|A = [-d\phi \wedge (\theta_1|A) + (1-\phi)(d\theta_1|A)] + [d\phi \wedge (\theta_2|A) + \phi(d\theta_2|A)] = \eta|A + d\phi \wedge (-\omega) = \eta|A - \delta(\omega)|A.$$

On U' and V': $d\theta|U' = d\theta_1|U' = \eta|U' = \eta|U' - \delta(\omega)|U'$ and similarly on V', since $\delta(\omega)$ vanishes on $U' \cup V'$. Hence $d\theta = \eta - \delta(\omega)$. ■

Theorem 40.4

Let $n \geq 1$. Then

$$\dim H^k(\mathbb{R}^n - \mathbf{0}) = \begin{cases} 0 & \text{for } k \neq n-1, \\ 1 & \text{for } k = n-1. \end{cases}$$

Proof.

Step 1. $n = 1$. Let $A = \mathbb{R}^1 - 0 = A_0 \cup A_1$, with A_0 the negative and A_1 the positive reals. If ω is a closed k-form on A with $k > 0$, $\omega|A_0$ and $\omega|A_1$ are closed; A_0, A_1 are star-convex, so there are $(k-1)$-forms η_i on A_i with $d\eta_i = \omega|A_i$. Let $\eta = \eta_0$ on A_0 and η_1 on A_1; then η is C^∞ on A and $d\eta = \omega$. For $k = 0$: let $f_0 = 0$ on A_0 and $f_0 = 1$ on A_1. f_0 is a closed 0-form that is not constant, hence not exact. $\{f_0\}$ is a basis of $H^0(A)$: if f is a closed 0-form on A, $f|A_0 = c_0$ and $f|A_1 = c_1$ are constants, so $f = c_1 f_0 + c_0(1 - f_0)$, i.e. $f = (c_1 - c_0)f_0 + c_0$, and $\{f\} = (c_1 - c_0)\{f_0\}$.

Step 2. If B is open in $\mathbb{R}^n$, then $B \times \mathbb{R}$ is open in $\mathbb{R}^{n+1}$ and $\dim H^k(B) = \dim H^k(B \times \mathbb{R})$ for all k. Let $g : B \to B \times \mathbb{R}$, $g(\mathbf{x}) = (\mathbf{x}, 0)$, and $h : B \times \mathbb{R} \to B$, $h(\mathbf{x}, s) = \mathbf{x}$. Then $h \circ g$ is the identity of B, and $g \circ h$ is differentiably homotopic to the identity of $B \times \mathbb{R}$ via the straight-line homotopy $H((\mathbf{x}, s), t) = t(\mathbf{x}, s) + (1-t)(\mathbf{x}, 0) = (\mathbf{x}, st)$. Apply Theorem 40.1.

Step 3. Assume the theorem for n $(n \geq 1)$ and prove it for $n + 1$. Let

$$U = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \geq 0\}, \qquad V = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \leq 0\},$$

so U is $\mathbb{R}^{n+1}$ minus the half-line $\mathbf{0} \times \mathbb{H}^1$ and V is $\mathbb{R}^{n+1}$ minus the half-line $\mathbf{0} \times \mathbb{L}^1$. Then $A = U \cap V = \mathbb{R}^n \times \mathbb{R}$ minus the line $\mathbf{0} \times \mathbb{R}$, i.e. $A = (\mathbb{R}^n - \mathbf{0}) \times \mathbb{R}$, and $X = U \cup V = \mathbb{R}^{n+1} - \mathbf{0}$. U is star-convex

with respect to $\mathbf{p} = (0, \dots, 0, -1)$ and V with respect to $\mathbf{q} = (0, \dots, 0, 1)$. By Theorem 40.3, $H^0(X)$ is trivial and $\dim H^{k+1}(X) = \dim H^k(A)$ for $k \geq 0$. By Step 2, $\dim H^k(A) = \dim H^k(\mathbb{R}^n - \mathbf{0})$, which by the induction hypothesis is 0 if $k \neq n-1$ and 1 if $k = n-1$. So $\dim H^{k+1}(\mathbb{R}^{n+1} - \mathbf{0})$ is 0 if $k+1 \neq n$ and 1 if $k+1 = n$. ■

Restated in terms of forms:

Theorem 40.5

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n \geq 1$.
(a) If $k \neq n-1$, then every closed k-form on A is exact on A.
(b) There is a closed $n-1$ form η_0 on A that is not exact. If η is any closed $n-1$ form on A, then there is a unique scalar c such that $\eta - c\eta_0$ is exact.

Proof [Supplement]. (a) $H^k(A) = 0$.

(b) $H^{n-1}(A)$ has dimension 1; take η_0 with $\{\eta_0\}$ a basis. Then $\{\eta\} = c\{\eta_0\}$ for a unique c. ■

This guarantees a closed non-exact $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ without giving a formula. An explicit one is

$$\eta_0 = \sum_{i=1}^{n} (-1)^{i-1} f_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \qquad f_i(\mathbf{x}) = x_i / \|\mathbf{x}\|^n.$$

Why η_0 is closed but not exact [Supplement]

$\eta_0 = \beta_{n-1} G$ for the vector field $G(\mathbf{x}) = (\mathbf{x}; \mathbf{x}/\|\mathbf{x}\|^n)$, so $d\eta_0 = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$ (Theorem 31.1). With $r = \|\mathbf{x}\|$, $D_i(x_i/r^n) = 1/r^n - n x_i^2/r^{n+2}$, so $\operatorname{div} G = n/r^n - n r^2/r^{n+2} = 0$. On $S^{n-1} = \partial B^n$ with the induced orientation, the corresponding normal is the outward one $N(\mathbf{x}) = (\mathbf{x}; \mathbf{x})$ (Lemma 38.7), and $\langle G, N \rangle = \|\mathbf{x}\|^2/\|\mathbf{x}\|^n = 1$ there; by Lemma 38.5, $\int_{S^{n-1}} \eta_0 = \mathrm{v}(S^{n-1}) \neq 0$. If $\eta_0 = d\theta$, Stokes' theorem would give $\int_{S^{n-1}} \eta_0 = \int_{\partial S^{n-1}} \theta = 0$ (S^{n-1} has empty boundary).

Theorem 40.6

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n > 1$. If η is a closed $n-1$ form in A, then η is exact in A if and only if

$$\int_{S^{n-1}} \eta = 0.$$

Proof. If η is exact, its integral over S^{n-1} is 0 by Stokes' theorem. Conversely suppose the integral is zero. By Theorem 40.5 there is a unique c with $\eta - c\eta_0$ exact. Then, by Stokes' theorem, $\int_{S^{n-1}} \eta = c \int_{S^{n-1}} \eta_0$. Since $\int_{S^{n-1}} \eta_0 \neq 0$, $c = 0$, and η is exact. ■